Robust Feature Learning for Satellite Image Classification under Adversarial Perturbations

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Abstract

Adversarial examples expose a fundamental vulnerability of deep neural networks, raising concerns for safety-critical applications such as satellite image classification. In this work, we study adversarial robustness on the EuroSAT dataset by comparing standard training, adversarial training with PGD, and mixed clean-adversarial training strategies. Inspired by Ilyas et al., Madry et al., and Xie et al., we analyze the trade-off between accuracy and robustness and investigate the role of adversarial examples in feature learning. The code used for this project can be found on the following GitHub page: https://github.com/TurboClem/Adversarial_examples

1 Introduction

Deep learning models, and in particular convolutional neural networks (CNNs), have become the dominant approach for image classification tasks. Their success extends to satellite image analysis, where they are used for land-use and land-cover classification, enabling applications such as environmental monitoring, urban planning, and climate modeling. On benchmark datasets, these models often achieve very high predictive accuracy, suggesting strong generalization capabilities.

However, recent work has revealed that high accuracy alone is not sufficient to guarantee reliability. Deep neural networks are vulnerable to adversarial examples: small, carefully designed perturbations to the input that are often imperceptible to humans, yet can drastically alter model predictions. This phenomenon raises critical concerns, especially in safety- and decision-critical domains such as remote sensing, where erroneous predictions can have significant real-world consequences.

A central question therefore arises: why are highly accurate models so vulnerable, and what does this imply for their deployment and training?

1.1 Non-Robust Features and the Source of Vulnerability

Ilyas et al. [1] provide a key insight into the origin of adversarial vulnerability. Rather than attributing adversarial examples to model bugs or optimization failures, they argue that such vulnerabilities arise from the presence of non-robust features in the data. These features are highly predictive of the label, yet extremely sensitive to small perturbations and often imperceptible to humans.

In particular, Experiment 3.2 of their work demonstrates that models trained exclusively on non-robust features can still achieve surprisingly high accuracy. By constructing datasets in which robust, human-interpretable features are suppressed while non-robust features are preserved, they show that these fragile features alone are sufficient for successful prediction. This result implies that standard training procedures naturally exploit non-robust but predictive signals, as they are statistically useful for minimizing classification error.

This observation has important implications. Since non-robust features play a significant role in prediction, adversarial attacks that target these features effectively remove a substantial portion of the model's predictive information. As a result, even small perturbations can lead to large drops in accuracy. From this perspective, adversarial attacks are genuinely dangerous: they exploit a core dependency of the model, not an incidental weakness.

1.2 Defending Against Attacks: Adversarial Training

Given that adversarial vulnerability is rooted in the features learned by the model, a natural question is whether training procedures can be modified to favor more robust representations. Madry et al. [2] propose adversarial training as a principled defense mechanism.

They formalize robust learning as a min–max optimization problem, in which the model is trained to minimize the worst-case loss within a bounded perturbation set. By incorporating strong adversarial attacks such as Projected Gradient Descent (PGD) during training, this approach explicitly discourages reliance on non-robust features and encourages the use of more stable, robust signals.

While adversarial training substantially improves robustness to worst-case perturbations, it comes with notable drawbacks. Most prominently, it introduces a trade-off between robustness and standard accuracy: models trained to be robust often perform worse on clean, unperturbed data. Moreover, adversarial training is computationally expensive and sensitive to the choice of threat model and hyperparameters.

1.3 Beyond the Trade-Off: Adversarial Examples as a Benefit

Interestingly, the interpretation of adversarial examples as purely harmful has been challenged by Xie et al. [3]. Their work suggests that adversarial examples can act as a powerful form of data augmentation rather than merely a defensive tool.

By mixing clean and adversarial examples during training, they demonstrate improvements not only in robustness but also in standard accuracy. This finding indicates that adversarial perturbations may help models discover more generalizable decision boundaries, partially mitigating the traditional accuracy–robustness trade-off observed in adversarial training.

Taken together, these works suggest a more nuanced view of adversarial examples: they expose dangerous dependencies on non-robust features, motivate robust optimization strategies, and may even enhance generalization when used appropriately.

1.4 Scope and Objectives of This Project

In this project, we study adversarial robustness and feature learning in the context of satellite image classification using the EuroSAT dataset. Motivated by the insights of Ilyas et al., Madry et al., and Xie et al., we aim to empirically analyze how different training strategies influence accuracy, robustness, and feature reliance.

We focus on three complementary paradigms:

- **Standard training**, which serves as a baseline and is expected to rely heavily on non-robust but predictive features.
- **Adversarial training** following Madry et al., which explicitly optimizes for worst-case robustness at the potential cost of clean accuracy.
- **Mixed clean and adversarial training** inspired by Xie et al., which explores whether adversarial examples can improve generalization rather than merely defend against attacks.

By evaluating these approaches within a unified experimental framework, we aim to better understand the role of non-robust features and the practical implications of adversarial robustness in satellite imagery.

2 Methodology and Studied Approaches

2.1 EuroSAT_RGB Dataset

EuroSAT_RGB is a land use and land cover (LULC) classification dataset based on Sentinel-2 satellite imagery covering 34 European countries. The dataset was introduced by Helber et al. in 2019 [4] and has become a benchmark for remote sensing image classification tasks. The RGB version of EuroSAT contains 27,000 labeled images across 10 distinct land cover classes. The original 10m resolution images were divided into 64×64 pixel patches, resulting in ground coverage of approximately 640×640 meters per image. For experimental consistency, the dataset is divided into training and test sets (80% train, 20% test).

Below is some samples of the dataset

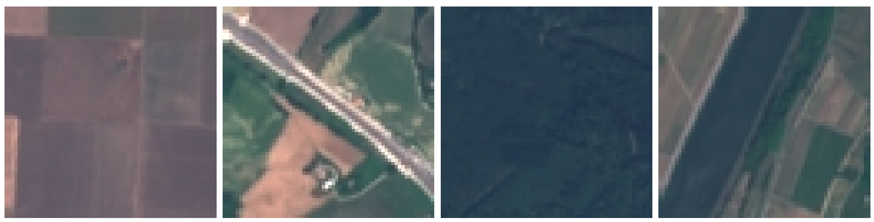

Figure 1: Example images from each class in EuroSAT_RGB dataset. Left to right: Annual Crop, Highway, Forest, River.

Despite its popularity, the dataset presents several inherent limitations that must be taken into account when interpreting experimental results. First, there exists substantial intra-class variation, as images belonging to the same class can exhibit significant visual diversity, such as different forest types or heterogeneous residential areas. At the same time, certain classes suffer from strong inter-class similarity, for example between annual crops and permanent crops, which can make class boundaries ambiguous even for human observers. Additionally, the dataset includes images captured across different seasons, introducing variations in color, texture, and illumination that are not directly related to semantic class identity. Finally, the fixed spatial resolution and patch size limit the amount of contextual information available to the model, resulting in scale ambiguity where local visual cues may be insufficient to reliably infer high-level land-use categories.

The dataset is publicly available under Creative Commons licenses and can be downloaded from:

- GitHub repository: `https://github.com/phelber/eurosat`
- Direct download link: `https://madm.dfki.de/files/sentinel/EuroSAT.zip`

2.2 Adversarial Attack Methods: FGSM and PGD

To evaluate adversarial robustness and to generate adversarial training data, we rely on two widely used gradient-based attack methods: the Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD). Both attacks aim to construct adversarial examples by maximizing the model's loss under a bounded perturbation constraint.

The Fast Gradient Sign Method (FGSM), introduced by Goodfellow et al. [5], is a single-step adversarial attack designed to efficiently generate adversarial examples. Given an input image x, its true label y, and a trained classifier f_θ, FGSM computes an adversarial perturbation by taking a step in the direction of the sign of the gradient of the loss with respect to the input:

$$x_{adv} = x + \epsilon \cdot sign\left(\nabla_x \mathcal{L}(f_\theta(x), y)\right),$$

where $\epsilon > 0$ controls the maximum perturbation magnitude under the ℓ_∞ norm constraint.

FGSM can be interpreted as a first-order linear approximation of the loss function around the input. While computationally efficient, FGSM often produces weaker adversarial examples compared to iterative attacks, and models trained solely against FGSM may remain vulnerable to stronger attacks.

Projected Gradient Descent (PGD) is an iterative extension of FGSM and is widely considered one of the strongest first-order adversarial attacks. PGD constructs adversarial examples by repeatedly applying small FGSM-like updates, followed by a projection step that ensures the perturbation remains within the allowed norm ball.

Starting from an initial point $x^{(0)} = x + \delta^{(0)}$, where $\delta^{(0)}$ is typically chosen at random within the ℓ_∞-ball, PGD performs the following update for $t = 0, \ldots, T-1$:

$$x^{(t+1)} = \Pi_{\mathcal{B}_\epsilon(x)} \left(x^{(t)} + \alpha \cdot sign \left(\nabla_x \mathcal{L}(f_\theta(x^{(t)}), y) \right) \right),$$

where α is the step size and $\Pi_{\mathcal{B}_\epsilon(x)}$ denotes projection onto the ℓ_∞-ball of radius ϵ centered at x.

By iteratively refining the perturbation, PGD is able to escape local linear regions of the loss landscape and find stronger adversarial examples than FGSM. For this reason, PGD is commonly used both as an evaluation attack and as the inner maximization procedure in adversarial training.

2.3 Baseline Model: Standard Image Classification

We begin by training a standard image classification model to establish a baseline in terms of predictive performance and robustness. We use a ResNet-18 architecture, a widely adopted convolutional neural network that employs residual connections to enable stable training of deep models.

Given an input image $x \in [0,1]^d$ and its label $y \in \{1, \ldots, K\}$, the model defines a classifier $f_\theta(x)$ parameterized by weights θ. Training is performed via empirical risk minimization:

$$\min_\theta \mathbb{E}_{(x,y)\sim\mathcal{D}} \left[\mathcal{L}(f_\theta(x), y) \right],$$

where $\mathcal{L}$ denotes the cross-entropy loss. This baseline is expected to achieve high accuracy on clean test data, but to be highly vulnerable to adversarial perturbations. It serves as a reference point against which all subsequent methods are evaluated.

2.4 Non-Robust Features as Predictive Signals (Ilyas et al.)

Ilyas et al. [1] go beyond a purely conceptual discussion and propose a concrete experimental protocol to isolate non-robust features. In Experiment 3.2, they explicitly construct a dataset in which the labels are intentionally *misassigned*.

Starting from a clean dataset $\{(x, y)\}$, a new target label $y' \neq y$ is sampled uniformly at random for each image. An adversarial perturbation (PGD for example) is then computed in the direction of this target label, such that the perturbed image is classified as y' by the baseline model:

$$x_{adv} = x + \delta, \quad \|\delta\|_\infty \leq \epsilon, \quad f_\theta(x_{adv}) = y'.$$

Crucially, the adversarial image x_{adv} is then assigned the label y', even though it remains visually indistinguishable from the original image x and does not contain robust, human-interpretable evidence of the new class.

The intuition behind this construction is the following. By attacking the image toward a randomly chosen incorrect class and relabeling it accordingly, the resulting dataset encodes label information almost exclusively through non-robust features introduced by the adversarial perturbation. Robust features, which align with human perception, are either inconsistent with the assigned label or actively misleading. As a result, any model trained successfully on this dataset must rely predominantly on non-robust features to make predictions.

If a classifier trained from scratch on this adversarially constructed, mislabeled dataset is able to achieve high accuracy on the test set, this provides direct empirical evidence that non-robust features alone are sufficient for prediction. Ilyas et al. observe exactly this behavior, demonstrating that non-robust features are not merely artifacts of adversarial attacks, but constitute a genuine and exploitable predictive signal.

In our work, we replicate this experimental setup by comparing models trained on such label-consistent adversarial datasets with models trained on randomly perturbed datasets. While both introduce perturbations, only adversarial perturbations are aligned with the (intentionally reassigned) labels, allowing us to distinguish meaningful non-robust features from uninformative noise.

2.5 Adversarial Training as a Defense (Madry et al.)

Given that vulnerability arises from reliance on non-robust features, a natural defense is to modify the training objective to penalize worst-case behavior. Madry et al. [2] propose adversarial training, which frames learning as a robust optimization problem.

The training objective is given by:

$$\min_{\theta} \mathbb{E}_{(x,y)\sim\mathcal{D}} \left[\max_{\|\delta\|_\infty \leq \epsilon} \mathcal{L}(f_\theta(x+\delta), y) \right].$$

The inner maximization is approximated using iterative attacks such as PGD.

This formulation can be interpreted as training the model against a worst-case adversary. The inner maximization corresponds to an attacker who is allowed to choose, for each input, the most damaging perturbation within the allowed threat model. The outer minimization then seeks model parameters that minimize the loss even under this strongest possible attack.

In this sense, adversarial training simulates a game between the classifier and an adversary: the attacker attempts to maximally degrade the model's performance, while the defender learns parameters that are robust to such perturbations. By optimizing for worst-case behavior rather than average-case performance, the model is encouraged to rely less on non-robust features that can be easily manipulated by the attacker.

While adversarial training significantly improves robustness to bounded perturbations, it often leads to a reduction in clean accuracy. This reflects an inherent trade-off between accuracy and robustness, which we explicitly measure in our experiments by evaluating performance on both clean and adversarially perturbed test sets.

2.6 Adversarial Examples as Data Augmentation (Xie et al.)

Adversarial Propagation (AdvProp), introduced by Xie et al. [3], offers an innovative perspective on adversarial examples: rather than viewing them solely as threats to be defended against, the approach leverages them as a sophisticated form of data augmentation that can simultaneously improve both robustness and clean accuracy.

Unlike traditional adversarial training which seeks to minimize loss on worst-case examples (a min-max approach), AdvProp treats adversarial examples as enriched variants of training data. The goal is to learn representations that perform well on both clean images and their perturbed versions, treating these two types as complementary views of the same instance.

Let x be an input image and y its label. We generate an adversarial example x_{adv} via a PGD attack. Instead of simply training on x_{adv} to minimize adversarial loss, AdvProp employs a *two-pass* strategy:

$$L_{total} = L(f_\theta(x), y) + L(f_\theta(x_{adv}), y) \tag{1}$$

where f_θ is the model. However, this simple formulation hides AdvProp's key innovation.

The fundamental problem with using adversarial examples in training is that their statistics (mean, variance) systematically differ from those of clean images. Using the same normalization layers (BatchNorm) for both types of images creates destructive interference: statistics estimated on adversarial images "pollute" those from clean images, degrading accuracy.

AdvProp solves this problem by introducing auxiliary BatchNorms. Specifically, the algorithm uses a dual-branch architecture; the model has two sets of BatchNorm layers which are respectively Main BatchNorms for clean images and Auxiliary BatchNorms for adversarial images. During training, clean images pass through main BatchNorms; adversarial images pass through auxiliary BatchNorms and convolutional weights are shared between both branches Unified evaluation: At inference, only the main BatchNorms are used, ensuring the final model is not burdened by additional components.

This approach offers several advantages over classical adversarial training. By isolating adversarial image statistics through the use of auxiliary Batch Normalization layers, AdvProp preserves the quality of representations learned from clean data and avoids the degradation of clean accuracy commonly observed in robust training methods. At the same time, the ability to model distinct data distributions allows the network to learn richer and more diverse feature representations. As a result, AdvProp mitigates the traditional accuracy–robustness trade-off, making it possible to improve robustness while maintaining or even enhancing performance on clean data.

AdvProp uses PGD with a moderate perturbation budget ϵ (e.g., 0.03 to 0.2 in our case). However, unlike Madry's adversarial training, the goal is not to find worst-case examples but to generate relevant variations of training images. We compare its performance with the standard training (baseline), but also Madry's adversarial training. We specifically evaluate the impact on accuracy over both clean and attacked datasets.

The complete algorithm can be summarized as follows:

Algorithm 1 Pseudo code of AdvProp

Data: A set of clean images with labels;
Result: Network parameter θ;
for *each training step* **do**
  Sample a clean image mini-batch x^c with label y
  Generate the corresponding adversarial mini-batch x^a using the auxiliary BNs
  Compute loss $L^c(\theta, x^c, y)$ on clean mini-batch x^c using the main BNs
  Compute loss $L^c(\theta, x^a, y)$ on adversarial mini-batch x^a using the auxiliary BNs
  Minimize the total loss w.r.t. network parameter θ: $\arg\min_\theta L^c(\theta, x^c, y) + L^c(\theta, x^a, y)$.
return θ

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3 Empirical Analysis

3.1 Baseline results

Our baseline model is a ResNet-18 architecture trained using standard empirical risk minimization on 80% of the EuroSAT dataset, with the remaining data split into the test set. A validation set is used for early stopping in order to prevent overfitting. The model is trained for a maximum of 50 epochs with an initial learning rate of 0.001, and a learning rate scheduler that reduces the learning rate on plateau. Additionally, we apply random rotations to the input images as a data augmentation strategy to improve the model's invariance to orientation changes and enhance its generalization capability. This training procedure yields excellent performance, with a clean test accuracy of **98.17%**.

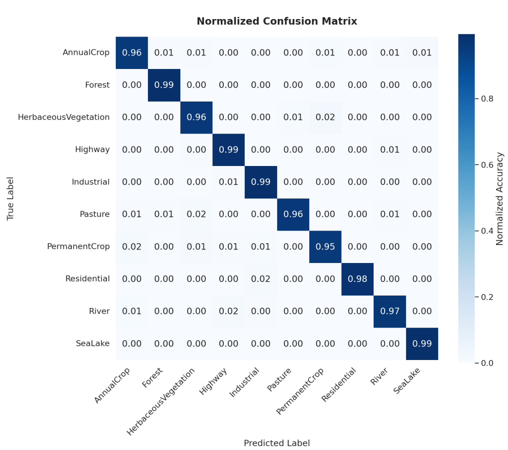

Figure 2: Normalized confusion matrix on the clean test set.

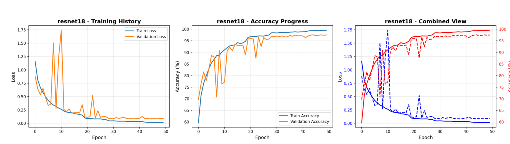

Figure 3: Training and validation accuracy and loss over epochs.

The normalized confusion matrix in Figure 2 shows that the model performs consistently well across all classes, with very few systematic confusions. This indicates that the learned representations are highly discriminative under standard, clean conditions. The training curves in Figure 3 further confirm stable convergence, with no significant gap between training and validation performance, suggesting that overfitting is effectively controlled.

Using this trained baseline model, we then evaluate adversarial robustness by constructing an adversarial test set from the clean test images. Adversarial examples are generated using a PGD attack with perturbation $\epsilon = 0.02$, step size $\alpha = 0.004$, and 5 iterations. When evaluated on this adversarially perturbed test set, the accuracy of the baseline model drops to approximately **20.28%**. This decrease proves that, despite its excellent clean performance, the standardly trained ResNet-18 is highly vulnerable to adversarial attacks. These results confirm that models trained without robustness considerations, and now we aim to understand why.

3.2 Non-Robust Features as Predictive Signals (Ilyas et al.)

We must now turn our attention to the nature of the adversarial attacks, and more specifically answer the question: do they simply produce noise, or do they attack non-robust features ?

We constructed an adversarial dataset by applying a PGD attack ($\epsilon = 0.3$, $\alpha = 0.075$, *20 steps*) to each image, minimizing the loss with respect to the (mislabeled) new class. The resulting perturbations are designed to amplify non-robust features. Our parameters result in a very strong adversarial attack, to the point where the true label is barely recognizable. Training on this adversarial dataset achieves nearly perfect performance: approximately 99% accuracy on the training, validation, and test sets.

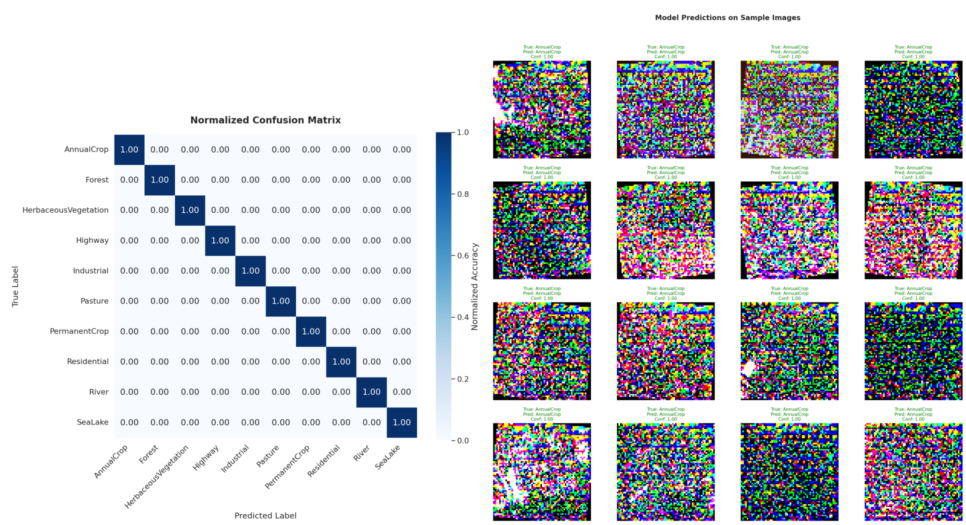

Figure 4: Results on the adversarial test set: normalized confusion matrix and sample predictions.

The primary objective of this experiment is to evaluate the model trained on adversarial examples on the original, clean dataset. A priori, it is unclear whether this model would outperform one trained solely on mislabeled but non-adversarial data, as our model has been exposed to both mislabeling and adversarial perturbations.
If the adversarial perturbations correspond only to random features (ie non features), performance on the clean test set should be comparable to that of the model trained on noisy dataset. If the adversarial perturbations correspond to non random features (thus still non-robust features), then our model trained on attacked dataset should yield better performance than the model trained on noisy dataset. However, the clean-test accuracy of the adversarially-trained model is 12%, compared to only 4% for the model trained on mislabeled noise. This significant gap provides strong empirical evidence for the existence and predictive utility of **non-robust features**.

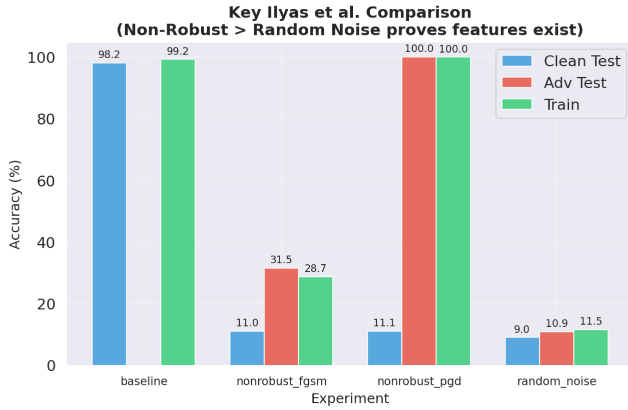

Figure 5: Comparison of models trained under different data conditions: FGSM, PGD, and random noise perturbations. Evaluations are shown for the adversarial training set, the adversarial test set, and the original clean test set.

As shown in the figure 5, training on the PGD-attacked dataset achieves perfect accuracy, indicating that the model relies exclusively on non-robust features exaggerated by the attack. Performance drops considerably on the clean test set because these **non-robust features are less prominent in natural images**. The FGSM attack, being a single-step approximation of PGD, produces milder perturbations, and yet performs on the dataset similarly to PGD trained model.

Notably, the adversarially-trained model still achieves **non-trivial accuracy** on clean data—roughly **2% higher** than the noise-trained model. This result further supports the conclusion that adversarial perturbations can uncover predictive, yet non-robust, patterns that retain some relevance even in the absence of an adversary.

3.3 Adversarial Training as a Defense (Madry et al.)

To evaluate adversarial training as a defense mechanism, we train a ResNet-18 in the same way as the baseline, but adversarial examples are generated on-the-fly using a strong PGD attack applied to the training images. The adversary is constructed using the baseline model to guide the perturbations, ensuring that the generated attacks are effective. Ideally, a robust model should maintain both strong robustness and acceptable clean accuracy.

We experiment with several PGD configurations by varying the perturbation ϵ, the step size α, and the number of iterations. The attack used during training is chosen to be stronger than the one used for to build the attacked test set with the baseline model, to avoid unfair evaluation scenarios in which the model is tested against unseen or stronger adversaries. Due to the high computational cost of adversarial training and limited GPU resources, the number of configurations and training runs we could explore remains constrained. Despite these efforts, the resulting performance falls short of our initial expectations. The results are presented in Table 1 for the two bests combinations.

Table 1: Clean accuracy and adversarial robustness under PGD attacks for adversarially trained models.

Model	ϵ	α	**Iterations**	**Accuracy (%)**	**Robustness (%)**
Baseline	Nan	Nan	Nan	**98.17**	20.28
Madry-1	0.01	0.002	7	36.19	24.43
Madry-2	0.03	0.008	7	33.78	**28.06**

Nevertheless, these results clearly illustrate the expected accuracy–robustness trade-off. We observe that increasing ϵ, and thus the strength of the adversarial attack, leads to improved robustness at the expense of clean accuracy. This behavior is expected, as training against stronger adversaries forces the model to rely on more conservative and robust features, which often reduces its ability to fit fine-grained, non-robust patterns that are beneficial for clean classification.

While the robustness improvement confirms the effectiveness of this framework in reducing vulnerability to adversarial perturbations, the limited clean generalization highlights the practical challenges of adversarial training. Indeed, these results do not come with theoretical robustness guarantees. Madry et al. themselves report several failure modes in which adversarial training does not converge to a meaningful robust solution. One such failure case we encountered is when the model collapses to predicting a single class for most inputs. This behavior can be interpreted as a degenerate solution to the min–max optimization objective, in which the model reduces worst-case loss by suppressing output variability rather than by learning robust and discriminative features.

3.4 Adversarial Examples as Data Augmentation (Xie et al.)

To evaluate the AdvProp approach proposed by Xie et al., we adapted our ResNet-18 implementation by incorporating the auxiliary BatchNorm layers required by the method. Training is performed on both clean images and their adversarially perturbed versions generated by a moderate PGD attack ($\epsilon = 0.02$, 7 iterations). Adversarial images pass through auxiliary BatchNorm layers while clean images use the main BatchNorm, thereby preventing the corruption of normalization statistics and preserving accuracy on unperturbed data.

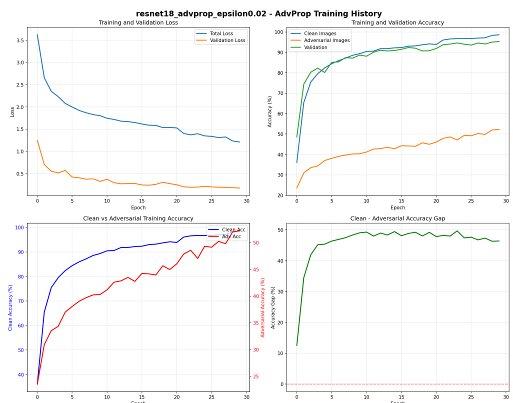

Figure 6: AdvProp learning curves (accuracy on training and validation sets).

The training dynamics (fig.6), show stable convergence, with validation accuracy reaching 95% after 30 epochs. This indicates that the model learns effectively from the mixed clean–adversarial batches. The model's evaluation on the original clean test set revealed a 95.67% accuracy, which is slightly under the baseline, possibly due to a smaller number of epochs (30 for 50 for the baseline).

On the adversarial test set (PGD with $\epsilon = 0.02$), the model achieved 38.02% accuracy, which represents a significant improvement over the baseline (20.28%) and even surpasses the robustness of the Madry-style adversarial training (28.06%), but is still quite low compared to the training performance. A likely explanation is that during test, the model incorrectly used the auxiliary BatchNorm statistics (meant for adversarial images) when processing.

4 Conclusion

In this project, we empirically studied adversarial examples on the EuroSAT dataset. We first showed that a standard ResNet-18 baseline achieves high clean accuracy but is highly vulnerable to adversarial attacks, due to its reliance on predictive but non-robust features, as highlighted by our experiment following Ilyas et al. We then implemented adversarial training following Madry et al., observing the expected trade-off between robustness and clean accuracy. Finally, we explored using adversarial examples as data augmentation, following Xie et al., to show their potential to improve generalization. All experiments were coded from scratch, providing a clean and complete pipeline. This project gave us a deeper understanding of adversarial examples, their causes, and defenses in practice. One limitation of our study is that we only considered a single type of adversarial attack, leaving exploration of other attack methods as a potential direction for future research.

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